

DEEP-DREAM: A TEMPORAL CONCEPT GRAPH FOR AGENTIC LONG-TERM MEMORY

Anonymous authors

Paper under double-blind review

ABSTRACT

Long-term memory is difficult because a growing history must stay easy to search without losing the exact wording needed for a later answer. Deep-Dream separates these jobs: original documents and exact passages are the factual record; concepts, relations, observations, and version links narrow the search; at query time, an agent opens and checks only the source text needed for the question. If an early mention does not identify which entity it refers to, the system keeps multiple candidates, and later observations can redirect the links without rewriting the source. We evaluate on MemoryAgentBench, LongMemEval, LoCoMo, and MEME. On MemoryAgentBench, under one model and one scorer, answering from a single retrieval result scores 0.333, answering through read-only memory tools 0.400, and answering from source text the agent has actually opened 0.710, with the margin concentrated on multi-mention and update questions; a same-model full-context reference bounds what selective reading still misses. Published Mem0 and Zep scores on LoCoMo and LongMemEval use different judges and rubric versions and serve as background, not controlled comparisons. In paired runs, the revised write implementation, which records observations first and merges compatible identities only after more context is available, reduces write requests per document by 46–65% at similar sampled quality.

1 INTRODUCTION

Consider choosing a restaurant for a friend. Months ago, she said, “I do not eat seafood, and I am cutting back on spicy food for now.” Last week, she added, “I am eating spicy food again.” Recalling only “dietary restrictions” loses which preference changed; remembering only the latest message loses the seafood restriction. A useful recollection connects both conversations to the same person, distinguishes what changed from what still applies, and preserves qualifiers such as “for now” and “cutting back.”

An agent faces the same problem across a growing history. It needs a compact way to locate related episodes without discarding the details that determine a future answer. A fixed summary may omit such details before the question is known; rereading the entire history makes each query pay for irrelevant context. Deep-Dream separates finding relevant memories from reading the evidence needed to answer.

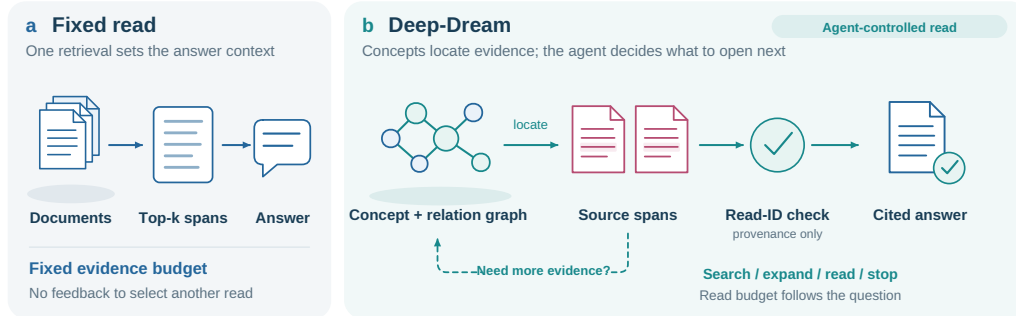


Figure 1: Separating navigation from evidence. Deep-Dream first locates candidate source text and then lets the agent read, verify, or stop within a budget.

We define Deep-Dream as a *document-first temporal concept graph for agentic long-term memory*. The definition has three parts.

- **Documents are the authority.** Original files, episodes, and exact source spans are retained. Concepts and relations point to this material; they do not replace it with a fact that the system is allowed to treat as true.
- **Observation time is the state axis.** Each processed input adds an observation and, when needed, a new version. Earlier observations can still be found. Event dates may be part of the observed text, but they do not create a second graph state timeline.
- **The graph is a navigation layer.** Stable concept families, relation edges, and version links narrow the search space. They answer “where should the agent look next?”; the source span answers “what was actually written?”

The same separation shapes construction and use. At write time, a low-confidence match is kept as a provisional family instead of forcing an irreversible merge. For example, an early note may say only “Alex approved the contract.” If the corpus contains a lab Alex and a vendor Alex, the system keeps two candidate families. Later notes such as “Alex from the lab presented with Dr. Liu” and “the vendor Alex renewed the account” add organization and relation evidence, so compatible pointers can be redirected without editing either source span. An eager merge would make a query about one Alex retrieve the other person’s contract and project facts. At read time, the agent can follow a concept, expand a relation neighborhood, open an episode, and return to the source span. A citation check accepts only source identifiers returned by the current read path.

The two write implementations differ in when an entity identity becomes final. The initial implementation (Base) commits an identity as each window arrives. The revised implementation (Align+) first records the observation and candidate matches, then merges compatible identities after the surrounding evidence has been processed. Both build the same kind of graph.

These design choices yield five predictions. Step-by-step source reading should help when an answer requires several mentions or a later correction. Observation times and version links should keep earlier and later evidence available together. Concept and relation links should connect evidence spread across episodes for multi-hop questions. An append-only store should fail when a stored fact must be deleted or treated as absent. Delaying identity decisions should reduce duplicate families and ingest cost, even if it does not improve every accuracy score.

Four benchmarks target distinct abilities. MemoryAgentBench keeps the questions and memory collection fixed while comparing one fixed retrieval result, repeated calls to read-only memory tools, and direct reading of source files with citation checks. LongMemEval writes complete conversations before testing whether an older statement and a later correction can be found together. LoCoMo reports binary J , the average of 1/0 correctness judgments, for single-hop, multi-hop, temporal, and open-domain questions. MEME tests deleted facts, facts linked to deleted items, and facts that were never present. Matched runs compare the two write implementations.

The paper makes four contributions:

- **A memory representation.** A document-first memory state $M_t = (F_t, O_t, G_t)$ separates stable concept families, append-only observations, and relation/version links.
- **A step-by-step reading method.** An agent controls concept search, relation expansion, source reading, verification, and stopping; the citation check ensures that cited text was actually opened.
- **An information-theoretic account.** Delayed identity alignment, the search/evidence decomposition, and query-time reading budgets are related to conditional information and rate-distortion, with assumptions and non-claims stated explicitly.
- **An evaluation across four memory tasks.** MAB, LongMemEval, LoCoMo, and MEME expose both strengths and failures across retrieval, temporal access, multi-hop linking, and deletion.

2 RELATED WORK

Memory benchmarks. LoCoMo and LongMemEval test long conversations, temporal updates, and multi-session reasoning (Maharana et al., 2024; Wu et al., 2025). MEME adds deletion, cascade, and absence checks (Jung et al., 2026). MemoryAgentBench (MAB) evaluates retrieval, test-time learning, long-range understanding, and selective forgetting in one official scoring framework (Hu et al., 2025); we use it as the primary instrument and keep its task-level metrics visible.

Structured memory. MemGPT, A-MEM, Mem0, HippoRAG-2, Cognee, MIRIX, and Zep/Graphiti improve persistence through summaries, notes, or graphs (Packer et al., 2023; Xu et al., 2025;

Chhikara et al., 2025; Gutiérrez et al., 2025; Markovic et al., 2025; Wang & Chen, 2025; Rasmussen et al., 2025). Deep-Dream shares the need for compact search but separates search records from answer evidence: a derived node points to a source span; it is not itself answer evidence.

Information-theoretic and associative memory. The information bottleneck formalizes a compact representation that preserves information relevant to a target variable (Tishby et al., 2000), while rate-distortion theory characterizes the cost of lossy reconstruction (Shannon, 1959). Associative-memory models explain content-addressable retrieval from partial cues (Ramsauer et al., 2021), and probabilistic record linkage separates false-link and missed-link errors (Fellegi & Sunter, 1969). We use these ideas as an analysis of the memory interface: the graph is a small search code, exact spans are the retained evidence channel, and delayed entity merges trade a tolerable missed-link cost against catastrophic false-link contamination.

Retrieval, long context, and source tracking. Retrieval-augmented generation selects evidence (Lewis et al., 2020). Agentic retrieval instead exposes the retrieval decision to the model: A-RAG hands keyword, semantic, and chunk-reading tools to a reasoning model over open-domain multi-hop corpora and reports accuracy that scales with test-time compute at comparable or fewer retrieved tokens (Du et al., 2026); its manual failure analysis identifies entity confusion as the dominant residual error, and dialogue and long-form memory remain outside its tested setting. Deep-Dream shares the progressive-disclosure principle but applies it to long-term memory rather than static corpora: concept and version search links, append-only observations, and source-span authority make later reads verifiable rather than merely cheaper, and entity confusion is the failure class that our alignment layer and delayed merges are designed to reduce. Long-context models can lose information when relevant spans are far from the attention window’s ends (Liu et al., 2024). Attribution work commonly checks citations after generation (Gao et al., 2023); our citation check only verifies that the cited source was opened. It does not prove entailment, and we report that limitation explicitly.

3 DEEP-DREAM

3.1 MEMORY MODEL

Let D be a source document and e_i an episode extracted from one of its spans. The persistent state is

$$M_t = (F_t, O_t, G_t),$$

where F_t contains stable concept families, O_t is an append-only store of observations with source spans, and G_t contains relation, neighborhood, and version links. Deep-Dream keeps D , episodes, versions, and exact source offsets as authoritative data. An entity family f is a search handle; each observation $o_{f,i}$ records what a source asserted, when it was observed, and where it can be reread. Relations use the same distinction between a derived link and the assertion that supports it. Active and historical observations remain distinguishable, so an update adds a pointer or version rather than overwriting source text.

The graph has one state axis: processing/observation time. An event date can remain in the source observation and can be used during retrieval, but it is not a second version clock. This choice keeps version order well-defined while preserving the temporal language that the document actually contains.

3.2 INGESTION AND ALIGNMENT: TWO ENGINE GENERATIONS

Base and Align+ differ in when entity identity is committed. They construct the same kind of graph over the same document representation. Base aligns and consolidates each window as it arrives, without a final reconciliation. Align+ separates evidence capture from identity commitment: windows are ingested concurrently, candidate matches remain open, and compatible families are merged only after all windows in the current task group have been processed. In plain terms, Base is “decide now”; Align+ is “record first, decide later with more context”.

The Remember pipeline performs heading-aware chunking, validated entity/relation extraction, conservative family alignment, batch candidate review, and final consistency pass. Over-splitting is recoverable because every observation keeps its source link. A later merge redirects compatible pointers; it does not rewrite or delete the observations that motivated the earlier split. A premature wrong merge is more dangerous because it makes two entities share one search entry before their evidence can be separated.

3.3 HOW ANSWERS OBTAIN EVIDENCE

All three methods use the same question stream, frozen libraries, scorer, answerer configuration, and task-group eligibility rule where the protocol permits. They differ only in how they obtain supporting text:

1. **Direct retrieval.** The system returns one fixed set of lexical, semantic, and graph matches. A separate answerer sees that set once and cannot ask for more.
2. **Tool-assisted reading.** The agent may call read-only search and record tools several times. The resulting records are then given to a separate answerer; this method does not open the original files.
3. **Source reading.** The system selects a relevant document group, the agent opens the original files with ordinary tools, and the answer cites only spans that it actually read.

Because tools, prompts, and compute change together, these are comparisons of complete systems rather than single-component interventions. A separate channel replay changes only the derived search channels.

A citation check runs without an LLM. It accepts a source identifier only when the current question’s tools returned it, it belongs to an eligible document group, and the system still has a pointer to the original text. This confirms that the citation came from text the agent opened; it does not guarantee that the text is relevant. The runtime records tool observations, source IDs, latency, exit status, and code/runtime hashes.

Candidate records combine lexical, semantic, concept, relation, and neighborhood channels by reciprocal-rank fusion and resolve to episodes and spans. In the source-reading method, the graph chooses where to read; it does not manufacture the content that is read.

3.4 PROGRESSIVE DISCLOSURE IN INFORMATION-THEORETIC TERMS

Write time: information is retained. Let $X_{1:t}$ be observations extracted from the first t source spans, Z their latent entity assignments, and $M_t = (F_t, O_t, G_t)$ the persistent state. Ingestion adds an observation and may redirect a family, but does not delete one:

$$\begin{aligned} O_{t+1} &= O_t \cup \{(X_{t+1}, \text{span}_{t+1})\}, \\ I(Z; O_{1:t+1}) &= I(Z; O_{1:t}) + I(Z; X_{t+1} \mid O_{1:t}) \geq I(Z; O_{1:t}). \end{aligned} \quad (1)$$

The equality is bookkeeping for the append-only observation stream, not a measured entropy gain. It explains why first-pass alignment can remain provisional: later evidence may refine the family partition while preserving the source material.

Read time: navigation plus source text. For a query q , let A be the compact search record returned by concept families and relation paths, E_C the exact spans opened from candidate set C , and Y the answer-relevant fact:

$$I(Y; A, E_C \mid Q = q) = I(Y; A \mid Q = q) + I(Y; E_C \mid A, Q = q). \quad (2)$$

The first term is the information used to find a candidate; the second is the additional fact information obtained by opening source text. A summary-only system replaces the source S by a write-time channel output $f(S)$ and may lose details needed by a later query. Deep-Dream keeps S available for later reading and lets the agent choose how much to read. This is an interpretation of the interface, not a claim that the implementation solves a variational bottleneck objective.

The agent chooses reading depth. Reading forms a chain $C_0(q) \subseteq C_1(q) \subseteq \dots \subseteq S$ under policy $a_{k+1} = \rho(q, M_t, C_{\leq k}, u_k)$, where u_k is current uncertainty. Every cited span belongs to S , so the citation check confirms that the cited text was opened; the remaining failure mode is that the relevant text was never found, in which a relevant span never enters C_k . If R_q denotes finding the relevant span and V_q denotes successful verification after reading it,

$$\Pr[\hat{Y} = Y \mid q] \geq \Pr[R_q \mid q] \Pr[V_q \mid R_q, q]. \quad (3)$$

This factorization predicts that the graph should help locate evidence, while additional source reading should matter most when the answer joins mentions, resolves an update, or checks a conflict.

4 EXPERIMENTS

The evaluation covers retrieval, temporal access, multi-hop linking, write efficiency, and deletion.

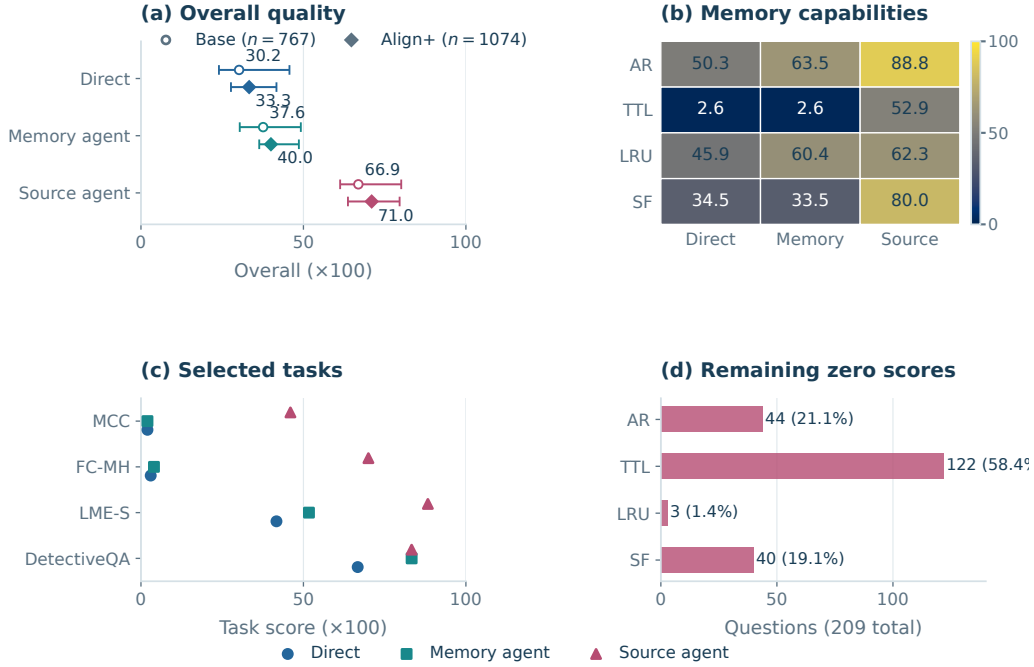


Figure 2: MemoryAgentBench results by method and task (official scorer, Kimi-K3 answerer). (a) Overall scores: Base on the frozen paired sample ($n=767$), Align+ on the extended sample ($n=1074$), with marginal 95% bootstrap intervals; (b) Align+ domain scores at the extended sample; (c) selected task scores; (d) all 209 residual zero-score items of the extended sample. The two sample definitions are labeled separately.

4.1 TEST SETUP AND TASK GROUPS

The primary MemoryAgentBench (MAB) evaluation has a paired 767-question sample over 10 task groups and 59 documents, plus an Align+ extension to 1,074 questions over 14 task groups. The paired sample supports Base–Align+ comparisons; the extension adds AR’s MH-QA and TTL’s Recommendation task groups and is evaluated only on Align+. All MAB scores use the official scorer¹ with Kimi-K3 as answerer. For LongMemEval, we first write the complete conversation set into the system and then use paired samples of $n = 25/25/27$ for the three answer methods. The separate 500-question LongMemEval-S run, the Mem0-compatible LoCoMo-1540 run, and MEME follow their published or stated test procedures. Detailed sampling and judge information is in Appendix A.3.

MAB reports percentages: 100 means every item in the row is correct and 0 means none, so higher is better. AR is *Accurate Retrieval*; TTL is *Test-Time Learning*; LRU is *Long-Range Understanding*; and SF is *Selective Forgetting*. MCC means *Multi-Class Classification*, Recom is movie recommendation scored by Recall@5, SH/MH mean single-/multi-hop, and FC-SH/FC-MH are single-/multi-hop FactConsolidation. Recall, J , exact-match, and Overall are higher-is-better measures; calls, tokens, and bytes are costs, so lower is better.

4.2 RQ1: DOES STEP-BY-STEP SOURCE READING IMPROVE ANSWERS?

Table 1 and Figure 2 compare the three MAB answer methods. On the extended Align+ sample, Overall is 0.333 for direct retrieval, 0.400 for the tool-assisted reading, and 0.710 for the source-reading agent. The paired Base sample gives 0.303, 0.376, and 0.669 for the same methods. The three methods change tools, prompts, and compute together, so this is a comparison of complete systems rather than a one-factor causal test.

The task pattern is more informative than Overall. Source reading raises the paired MCC and FC-MH rows from near-zero retrieval scores to 46 and 70, respectively. On the two extension task groups that exceed the full-context reference’s character budget, the source-reading method reaches 77.0 on MH-QA and 59.8 on Recommendation. The same-model full-context reference covers 614 of 1,074

¹The scorer version used for these runs is commit 455306d.

Task / domain	Direct retrieval		Tool-assisted reading		Source reading		Align+ gain
	Base	Align+	Base	Align+	Base	Align+	Align+ gain
AR: Accurate Retrieval							
SH-QA	54.0	48.0	84.0	72.0	94.0	<u>92.0</u>	44.0
MH-QA [‡]	–	22.0	–	<u>42.0</u>	–	77.0	–
LongMemEval-S*	26.7	41.7	43.3	51.7	90.0	<u>88.3</u>	46.6
EventQA	<u>90.3</u>	89.7	<u>90.7</u>	88.3	98.0	98.0	8.3
TTL: Temporal and Test-time Learning							
MCC, icl_clinic150	4.0	2.0	3.0	2.0	<u>43.0</u>	46.0	44.0
Recom [‡]	–	<u>3.2</u>	–	<u>3.2</u>	–	59.8	–
LRU: Long-range Understanding							
∞ Bench-Sum [†]	0.0	0.0	<u>28.4</u>	29.8	21.2	0.0	0.0
DetectiveQA	50.0	<u>66.7</u>	50.0	83.3	83.3	83.3	16.6
SF: Selective Forgetting							
FC-SH	<u>68.0</u>	66.0	67.0	63.0	90.0	90.0	24.0
FC-MH	2.0	3.0	4.0	4.0	<u>67.0</u>	70.0	67.0
Overall ($n = 767$)	30.3	32.4	37.6	40.7	66.9	<u>65.1</u>	32.7
Overall ($n=1074$, extended)	–	33.3	–	40.0	–	71.0	–

Table 1: MemoryAgentBench; scores are official component scores $\times 100$. Task rows and the Overall ($n=767$) row use the paired sample shared by both engines. [‡] marks the two task groups added when the Align+ engine was extended in place to $n=1074$ (RULER-421K multi-hop QA and the full Redial recommendation corpus); the Base run was frozen at the paired sample, and both task groups exceed the full-context row’s strict character budget (Appendix A.3), so they carry no same-model coverage. The Overall ($n=1074$) row is that extended sample; scores from the two samples are never subtracted. Bold/underlined entries are best/second-best within each task and sample, * marks sampled LongMemEval-S, and [†] marks the singleton ∞ Bench-Sum row of the paired sample (a second task group enters only the extended sample, so LRU remains directional: $n=7$ paired, 14 extended).

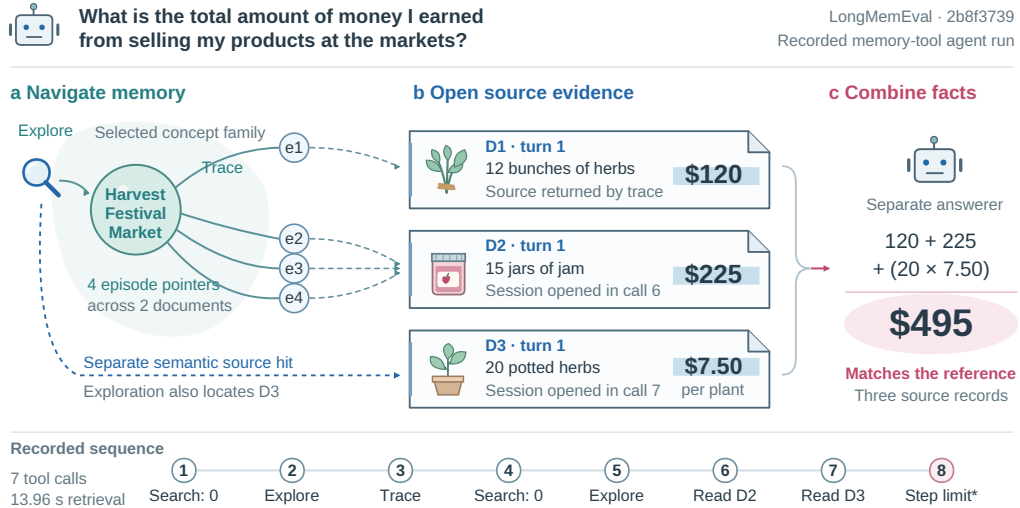


Figure 3: Recorded LongMemEval trace (question 2b8f3739): concept tracing exposes four episode pointers across two documents, and a separate semantic hit locates the third sales record. Seven successful calls take 13.96 s; step 8 reaches the limit after a malformed response, then a separate answerer computes $120 + 225 + (20 \times 7.5) = 495$.

questions and scores 78.7 on those covered questions; the source-reading method scores 70.8 on the same set. The reference is a reference under a strict budget, not a universal ceiling, and it has no score on the 460 questions whose document groups exceed that budget.

4.3 RQ2: DOES A TEMPORAL, VERSIONED STATE HELP UPDATES AND SESSIONS?

LongMemEval tests whether an earlier observation and a later correction remain jointly accessible. In paired runs that first write the full conversation set, direct retrieval moves from 0.68 to 0.72, the memory-tool path also moves from 0.68 to 0.72, and the source-reading method moves from 0.889 to 0.926. These tracks are small and directional. The independent 500-question LongMemEval-S run used Qwen3.7-plus for cross-judging and obtained 84.60%; it is listed separately because it used a different test procedure.

The source-reading method is strongest on update and temporal questions, while preference questions remain difficult because they require latent user-state modeling. The result concerns access to changing observations, not all forms of temporal reasoning.

4.4 RQ3: DO CONCEPT AND RELATION PATHS SUPPORT MULTI-HOP CONSOLIDATION?

We score LoCoMo-1540 with the binary CORRECT/WRONG rubric of the Mem0 benchmark repository pinned at commit `4b61c5d`—a 2026 revision whose date, duration, and list-tolerance rules differ from the 2025 paper’s Appendix A—and with Kimi-K3 as both answerer and judge; the same answers score 94.22% under an independent Qwen3.7-plus cross-judge (agreement 98.51%). Deep-Dream obtains 95.19% weighted J (percent judged correct), a single diagnostic run carrying a disclosed legacy-fingerprint caveat rather than a fingerprint-complete formal result. For context, reweighting Mem0’s published per-category means under the same 841/282/96/321 question counts gives ≈ 62.14 (its paper’s Table 2 reports Overall $J = 66.88$), and Zep’s reweighted categories give ≈ 56.32 ; those rows use different answerers, judges, and protocols and are background references, not controlled comparisons.

The category profile is consistent with the intended role of the graph: multi-hop and temporal questions benefit from connecting episodes and retaining update order, while open-domain questions are less dependent on a specific relation path.

4.5 RQ4: DOES PROGRESSIVE ALIGNMENT MAKE CONSTRUCTION REVISABLE AND ECONOMICAL?

Base commits identity window by window; Align+ defers compatible merges until all windows in the current task group have been processed. In paired ingest artifacts, calls fall from 609 to 329 per MAB document and from 63.8 to 22.0 per LongMemEval document. Same-name duplicate families fall from 340 to 124 on MAB and from 482 to 99 on LongMemEval. MCC and FC-MH each increase by 0.03, but their intervals include zero. Concurrency, convergence, retries, timeouts, and throughput fixes changed together, so this result supports a construction cost/revisability trade-off rather than a component-level causal claim.

4.6 RQ5: WHERE DOES THE APPEND-ONLY DESIGN FAIL?

MEME exposes the current boundary. Deletion accuracy is 0% on 100 episodes; cascade and absence real accuracy are 46.34% and 59.23%. The append-only store preserves prior observations and links, but the current query path has no deletion suppression or propagation rule. Temporal revision and deletion are therefore separate capabilities.

4.7 COMPARISON WITH OTHER SYSTEMS

Table 2 places Deep-Dream beside published MAB, LoCoMo, and LongMemEval-S results. Each block states its answerer, judge, subset, and test procedure. On MAB, the full-context reference and the source-reading method use the same model on their common subset. The MAB pair is a controlled comparison; rows that span protocols are placed side by side for context, and no cross-protocol difference is claimed.

4.8 MECHANISM DIAGNOSTICS: OFFLINE RETRIEVAL AND ONLINE READING

This diagnostic asks whether the graph improves the ranking before the agent opens source text. We replayed the same 210 questions with four index variants: raw documents plus keyword search (BM25) over episodes; semantic links added; graph neighbors added; and relation links added. Recall-any@10 asks whether at least one relevant item appears in the top ten; Recall-all@10 asks whether all relevant items appear; MRR@10 and nDCG@10 reward relevant items appearing earlier in the ranking. Recall-any@10 is 72.38, 72.38, 71.43, and 72.38 percent across the four variants;

(a) MemoryAgentBench (official scorer)							
System / answerer	AR	TTL	LRU	SF	Overall		
Full-context / GPT-4o	58.1	50.0	54.9	32.5	48.8		
Full-context / GPT-4o-mini	49.2	48.6	46.2	25.0	42.2		
Full-context / GPT-4.1-mini	71.8	46.2	49.1	20.5	46.9		
Full-context / Gemini-2.0-Flash	65.1	46.4	41.6	16.5	42.4		
Full-context / Claude-3.7-Sonnet	59.7	53.9	62.2	22.5	49.6		
BM25 RAG / GPT-4o-mini	60.5	44.5	35.6	25.5	41.5		
HippoRAG-2 / GPT-4o-mini	65.1	35.8	36.2	29.5	41.6		
MIRIX / GPT-4.1-mini	63.0	35.7	40.5	11.5	37.7		
MemGPT / GPT-4o-mini	34.3	40.8	22.4	15.5	28.3		
Zep / GPT-4o-mini	37.5	37.5	16.2	5.0	24.0		
Mem0 / GPT-4o-mini	32.6	21.2	20.7	10.0	21.1		
Deep-Dream / Full context (Kimi-K3, $n=614$)	96.8	75.0	65.2	78.0	78.7		
Deep-Dream / Direct retrieval (Kimi-K3)	50.3	2.6	45.9	34.5	33.3		
Deep-Dream / Memory-tool agent (Kimi-K3)	63.5	2.6	60.4	33.5	40.0		
Deep-Dream / Source reading (Kimi-K3)	88.8	52.9	62.3	80.0	71.0		
(b) LoCoMo-J score (%; Mem0 scoring)							
System / judge	Single-hop	Multi-hop	Open-domain	Temporal	Weighted		
Mem0 / GPT-4o-mini	67.13	51.15	72.93	55.51	≈ 62.14		
Mem0 ^g / GPT-4o-mini	65.71	47.19	75.71	58.13	≈ 61.36		
Zep / GPT-4o-mini	61.70	41.35	76.60	49.31	≈ 56.32		
LangMem / GPT-4o-mini	62.23	47.92	71.12	23.43	≈ 52.08		
OpenAI / GPT-4o-mini	63.79	42.92	62.29	21.71	≈ 51.10		
A-Mem* / GPT-4o-mini	39.79	18.85	54.05	49.91	≈ 38.95		
Deep-Dream / Kimi-K3	97.38	93.62	80.21	95.33	95.19		
Deep-Dream / Qwen3.7-plus cross-judge	96.91	92.91	77.08	93.46	94.22		
(c) LongMemEval-S (%)							
System / answerer	Overall	Sess. pref.	Sess. asst.	Temporal	Multi-sess.	Knowl. upd.	Sess. user
Full-context / GPT-4o-mini	55.4	30.0	81.8	36.5	40.6	76.9	81.4
Zep / GPT-4o-mini	63.8	53.3	75.0	54.1	47.4	74.4	92.9
Full-context / GPT-4o	60.2	20.0	94.6	45.1	44.3	78.2	81.4
Zep / GPT-4o	71.2	56.7	80.4	62.4	57.9	83.3	92.9
Deep-Dream / Qwen3.7-plus	84.6	53.3	100.0	84.2	74.4	91.0	98.6

Table 2: Comparison with published MemoryAgentBench, LoCoMo, and LongMemEval-S results. Each block lists its answerer, judge, subset, and test procedure; published rows are from Hu et al. (Hu et al., 2025), Chhikara et al. (Chhikara et al., 2025), and Rasmussen et al. (Rasmussen et al., 2025). In (b), the weighted column is recomputed from published category means under the 841/282/96/321 counts; cross-block rows use different protocols and are background, not controlled comparisons. Our MAB rows use the Align+ extended sample ($n=1074$, 14 task groups; Appendix A.3); the same-model full-context row is a reference at 614/1,074 coverage, rather than a competing system, and its margin over the source-reading agent (-7.9 points, 95% CI $[-11.9, -3.9]$) is analyzed in Section 4.2. LRU has only $n=14$ questions, including $n=2$ from ∞ Bench-Sum.

every paired difference lies within the ± 2.4 -point 95% bootstrap equivalence bound computed from the same replay, and the other metrics show the same pattern.

The offline replay therefore gives no evidence that adding graph links alone improves recall. Online, the graph still chooses document groups, versions, and relation links before the agent opens source text. The citation check accepts only source IDs that the agent actually received from a tool call. Figure 7 reports the cost of the three answer paths: source reading uses more calls and tokens while producing the strongest answers.

4.9 WHERE THE SYSTEM STILL FAILS

The source-reading method still gets 29.0% of MAB questions wrong; errors concentrate in temporal/test-time learning and fact consolidation. Direct retrieval and tool-assisted reading tie on the small paired LongMemEval tracks, the LRU estimate is based on only 14 questions, and the 210-question replay finds no measurable change in offline recall. MEME’s 0% deletion accuracy and the change caused by a single example show that preserving source text does not yet provide deletion or stable estimates for every task.

5 ANALYSIS AND LIMITATIONS

The scientific problem is how to make a growing record of experience searchable and trustworthy when a query can read only part of it. Summary-only memory loses wording and qualifications at write time; graph-only memory can mistake derived links for facts; fixed full-context prompting pays for every document and eventually exceeds the window. Deep-Dream separates three responsibilities: concepts and relations locate candidates, observations and source spans authorize facts, and the agent decides how much evidence to open.

This separation changes what a graph should be judged on. The graph need not contain a complete answer, and graph channels need not improve offline top-ten recall by themselves. Their role is to provide a compact route to document groups, versions, and related episodes before source reading. The channel replay supports this interpretation: the graph is an address layer, not a second fact store. MAB gains on questions that join mentions or check updates therefore support navigation plus source reading; LoCoMo adds the complementary case in which relation paths connect episodes while linked text remains the authority.

Progressive alignment addresses a different scientific question: how should identity uncertainty be handled when the first mention is ambiguous? Keeping two candidate families preserves a recoverable uncertainty; later observations can redirect pointers and merge compatible families without rewriting the observations that caused the split. An early wrong merge contaminates one address with two histories and is harder to undo. Align+ is therefore a revision policy, not merely a faster extractor. Its lower call counts and fewer duplicate families are consistent with less repeated work, but bundled concurrency and runtime changes prevent a component-level causal claim.

The temporal principle gives long-term memory a precise state axis. Processing/observation time orders versions; event dates remain content in the observations. New information adds a version or pointer while leaving the old span available, so an earlier statement and a later correction remain jointly searchable. This preserves the history needed to explain change, but it does not specify which observations should be suppressed after deletion. MEME exposes that missing operation: temporal revision and deletion are distinct capabilities.

At read time, the information-theoretic model predicts adaptive allocation rather than one fixed summary size (Section 3.4). The address term in Equation (2) is cheap and lossy enough to navigate; the source term supplies the wording needed for the answer. A local question can stop early, whereas a multi-hop, update, or conflict question should expand relations and read more spans. The answer decomposition in Equation (3) separates failure to find a span from failure to verify it. Current results support the value of source reading and revision-aware writing, but they do not measure entropy reduction directly.

The evidence has limits. MAB paths are complete-system comparisons, not single-component interventions; LongMemEval and LRU subsets are small; and published cross-system rows use different models, judges, and subsets. The equations describe intended information flow, not measured gains or an optimality proof. The citation gate confirms that an opened span was returned by a tool, not that it entails the answer. Future work should measure entity uncertainty, merge contamination, accuracy under a fixed read budget, deletion and as-of-time behavior, conflict handling, and confidence calibration.

6 CONCLUSION

Long-term memory must locate relevant history, preserve the conditions that make a fact trustworthy, and revise identities and beliefs when later observations arrive. Deep-Dream answers with a document-first temporal concept graph: concepts, relations, and versions form the address layer; append-only observations and source spans form the authority layer; and an agent controls evidence expansion.

This differs from summary memory, which commits information loss during writing; graph-only memory, which risks treating derived links as facts; and fixed full-context prompting, which spends the same large input budget on every query. The results show where the distinction matters: source reading helps multi-mention and update questions, relation links connect episodes, and deferred alignment reduces repeated ingestion work. They also show what remains open: graph links alone do not guarantee offline recall gains, deletion is not yet history suppression, and some temporal tasks remain difficult.

The broader contribution is a set of design questions for long-term memory: what is compressed for addressing, what remains authoritative, how identity is revised, and who controls the read budget. These questions provide a basis for future deletion, conflict, and evidence-calibration benchmarks.

AI ASSISTANCE

We used LLM-based assistance, including OpenAI Codex, to draft, restructure, and edit the manuscript; prepare a Chinese reading version; search for related literature and bibliographic information; and assist with analysis and plotting code. Generative image tools were used to create and revise explanatory framework figures.

AI assistance also contributed to the information-theoretic interpretation of the architecture, the formulation of mathematical claims and their supporting arguments, and the discussion of experimental methodology and results. This included the treatment of progressive entity alignment, search–evidence separation, and query-dependent reading budgets. These arguments are conditional on the assumptions stated in the paper; AI assistance does not constitute independent proof verification or empirical validation.

No synthetic evaluation dataset was created as part of this paper-preparation workflow. Reported benchmark scores are drawn from recorded experimental runs rather than generated performance estimates. The authors remain responsible for the final text, citations, mathematical claims, code, figures, and empirical conclusions, including verification of AI-assisted material.

REFERENCES

- Herman Chernoff. A measure of asymptotic efficiency for tests of a hypothesis based on the sum of observations. *The Annals of Mathematical Statistics*, 23(4):493–507, 1952. doi: 10.1214/aoms/1177729330. URL <https://doi.org/10.1214/aoms/1177729330>.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025. URL <https://arxiv.org/abs/2504.19413>.
- Mingxuan Du, Benfeng Xu, Chiwei Zhu, Shaohan Wang, Pengyu Wang, Xiaorui Wang, and Zhendong Mao. A-RAG: Scaling agentic retrieval-augmented generation via hierarchical retrieval interfaces. *arXiv preprint arXiv:2602.03442*, 2026. URL <https://arxiv.org/abs/2602.03442>.
- Ivan P. Fellegi and Alan B. Sunter. A theory for record linkage. *Journal of the American Statistical Association*, 64(328):1183–1210, 1969. doi: 10.1080/01621459.1969.10501049. URL <https://doi.org/10.1080/01621459.1969.10501049>.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 6465–6488, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.398. URL <https://aclanthology.org/2023.emnlp-main.398/>.
- Bernal Jiménez Gutiérrez, Yiheng Shu, Weijian Qi, Sizhe Zhou, and Yu Su. From rag to memory: Non-parametric continual learning for large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 21497–21515. PMLR, 2025. URL <https://proceedings.mlr.press/v267/gutierrez25a.html>.
- Yuanzhe Hu, Yu Wang, and Julian McAuley. Evaluating memory in llm agents via incremental multi-turn interactions. *arXiv preprint arXiv:2507.05257*, 2025. URL <https://arxiv.org/abs/2507.05257>.
- Seokwon Jung, Alexander Rubinstein, Arnas Uselis, Sangdoo Yun, and Seong Joon Oh. Meme: Multi-entity & evolving memory evaluation. *arXiv preprint arXiv:2605.12477*, 2026. URL <https://arxiv.org/abs/2605.12477>.
- Patrick S. H. Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pp. 9459–9474, 2020. URL <https://arxiv.org/abs/2005.11401>.
- Yifei Li, Weidong Guo, Lingling Zhang, Rongman Xu, Muye Huang, Hui Liu, Lijiao Xu, Yu Xu, and Jun Liu. Locomo-plus: Beyond-factual cognitive memory evaluation framework for LLM agents. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 25085–25100, San Diego, California, United States, July 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.acl-long.1150. URL <https://aclanthology.org/2026.acl-long.1150/>.

- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Parandekar, Danqi Chen, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. doi: 10.1162/tacl_a_00638. URL <https://arxiv.org/abs/2307.03172>.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 13851–13870, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.747. URL <https://aclanthology.org/2024.acl-long.747/>.
- Vasilije Markovic, Lazar Obradovic, Laszlo Hajdu, and Jovan Pavlovic. Optimizing the interface between knowledge graphs and llms for complex reasoning. *arXiv preprint arXiv:2505.24478*, 2025. URL <https://arxiv.org/abs/2505.24478>.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023. URL <https://arxiv.org/abs/2310.08560>.
- Hubert Ramsauer, Bernhard Schöfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Lukas Gruber, Markus Holzleitner, Thomas Adler, David Kreil, Michael K. Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2008.02217>.
- Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. Zep: A temporal knowledge graph architecture for agent memory. *arXiv preprint arXiv:2501.13956*, 2025. URL <https://arxiv.org/abs/2501.13956>.
- Claude E. Shannon. Coding theorems for a discrete source with a fidelity criterion. In *IRE National Convention Record*, volume 7, pp. 142–163, 1959. URL <https://ieeexplore.ieee.org/document/5311476>.
- Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. Alfworld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=0IOX0YcCdTn>. ICLR 2021.
- Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. *arXiv preprint physics/0004057*, 2000. URL <https://arxiv.org/abs/physics/0004057>.
- Yu Wang and Xi Chen. Mirix: Multi-agent memory system for llm-based agents. *arXiv preprint arXiv:2507.07957*, 2025. URL <https://arxiv.org/abs/2507.07957>.
- Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. Longmemeval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/d813d324dbf0598bbdc9c8e79740ed01-Abstract-Conference.html. ICLR 2025.
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-mem: Agentic memory for llm agents. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-0593. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/19909c36f51abc4856b4560aff3d36d6-Abstract-Conference.html.
- Terry Yue Zhuo, Minh Chien Vu, Jenny Chim, Han Hu, Wenhao Yu, Ratnadira Widyasari, et al. Bigcodebench: Benchmarking code generation with diverse function calls and complex instructions. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/a6a90bcc2aa470c3871b2d39a67d26e8-Abstract-Conference.html. ICLR 2025.

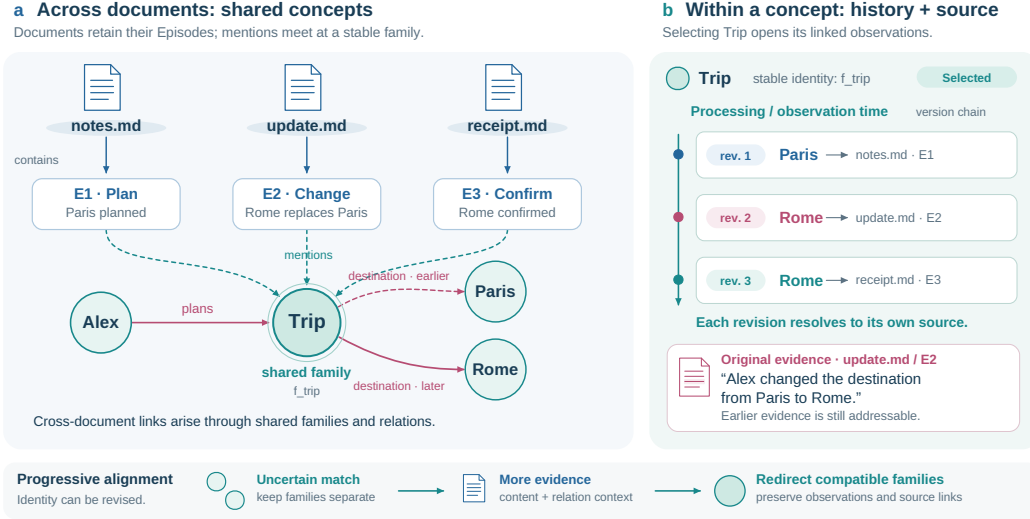


Figure 4: Deep-Dream’s document-first temporal concept graph. Multiple documents keep their Episodes and connect through shared concept families and relation edges. Selecting a family exposes its observation-time version chain and the exact source span for each revision; uncertain matches can remain separate and converge when later content and relation context arrive.

Property	Direct retrieval	Tool-assisted reading	Source reading
Source access	top- k spans	derived records	verbatim sources
Tools	none	memory tools	read, grep, script
Answerer	separate	separate	verifying agent
Loop	none	several read-only calls	several read-and-cite calls
Cost / question	27.5k tokens, 1.0 LLM calls	43.2k tokens, 6.5 LLM calls	328.6k tokens, 10.2 LLM calls

Table 3: Differences between the three reading methods. All three paths share frozen libraries, question stream, Kimi-K3, and the official scorer.

A REPRODUCIBILITY AND AUDIT DETAILS

A.1 ARTIFACT LEDGER

The machine-readable ledger is generated by the released ledger and recomputation scripts. It records the source summary, dataset, and per-item JSONL SHA-256 values for every run the paper cites, and is regenerated and compared byte-for-byte with the script output before paper writing. The current ledger covers the MemoryAgentBench tracks (two engine generations $\times$ three answer paths, official scorer at commit 455306d, identical `entity2id` fingerprint), LongMemEval complete-write pairs and the 500-question run, LoCoMo-1986/1540 and the K3-as-answerer diagnostic, LoCoMo-Plus-401 (Li et al., 2026), the X7 channel diagnostic, the source-expansion diagnostic, the repaired budget-frontier replay, MEME, BigCodeBench (Zhuo et al., 2025), and ALFWorld (Shridhar et al., 2021). Per-run manifests record dataset and prompt provenance, model identifiers, tool observations, evidence IDs, latency, exit status, and code/runtime hashes; failed integrity checks and negative results are retained rather than dropped. The ledger-generation and recompute scripts ship with the released supplementary material so the chain is regenerable end to end.

The released supplementary material contains the machine-readable manifests, per-item scores, prompts, and runtime hashes needed to regenerate every table. The paper reports dataset, model, judge, subset, and integrity status alongside each result so that the comparisons remain auditable without exposing machine-specific storage locations.

A.2 VERSION PROVENANCE AND ALIGNMENT A/B STUDY

The accompanying evidence package adds a complete three-arm alignment study on LongMemEval-S: three document groups (42+49+46 documents, 137 documents per arm), write all documents first, and the same Kimi-K3 endpoint. Arm A is the baseline without batch candidate review; B1 enables batch candidate review; B2 additionally enables the revised convergence implementation (parallel alignment, deferred non-destructive merges, and a final consistency check). The saved B1 and B2 service configurations are byte-identical; their difference is in the runtime code and the B2 switch, so

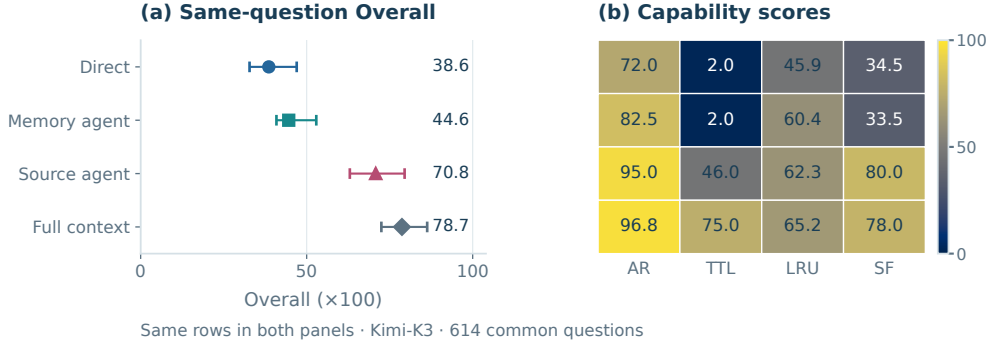


Figure 5: Same-question alignment on the $n=614$ questions covered by the same-model full-context reference (Kimi-K3 answerer; official macro scorer). (a) Overall scores with marginal 95% bootstrap intervals; the source-reading method scores below the reference by 7.9 points [95% CI $-11.9, -3.9$]. (b) Domain profiles: the reference’s margin concentrates in TTL, while the source-reading method matches or exceeds the reference on AR, LRU, and SF.

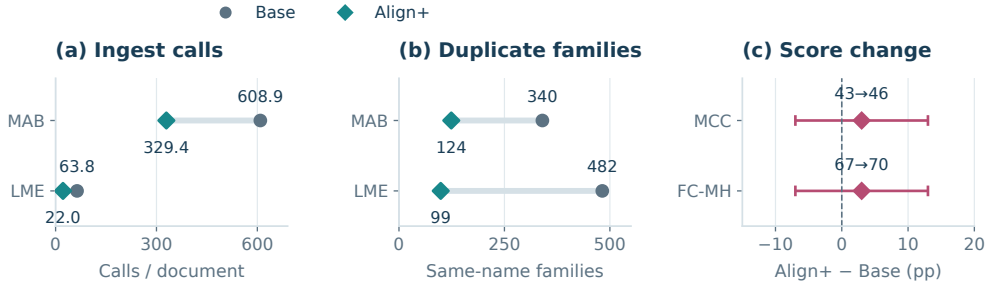


Figure 6: Paired construction diagnostics for the bundled Base→Align+ revision. (a) Ingest calls and same-name duplicate families decrease on the documented MAB and LongMemEval intersections. (b) The paired MCC and FC-MH slices each increase by 3 points; intervals are retained because this is a bundled comparison, not a component ablation.

the arm assignment is read from the manifest and code snapshot rather than inferred from filenames. The study is powered for efficiency and consolidation, not quality: each document group contributes only one question per evaluation track.

The separate LME Base/Align+ engine diff (initial/revised implementation) is a post-hoc reconstruction from commits `73d9c5e` to `8fa9651`. The latter also contains four throughput fixes, and therefore cannot by itself establish which fixes were active at each point in the Align+ run. We treat the manifest-embedded pipeline configuration, runtime-policy hash, and skill hash as authoritative; the reconstructed diff is provenance for the uncommitted code, not a claim that every later fix was active from the first revised run.

A.3 MEMORYAGENTBENCH SAMPLING FRAME AND CONTAMINATION NOTE

The sampled subset was drawn with a fixed task-group list recorded in the released ledger: a 767-question paired sample over 10 task groups and 59 documents, extended in place on the Align+ engine with four task groups (RULER-421K multi-hop QA, 100 questions; the full Redial recommendation corpus, 200 questions; one further DetectiveQA task group, 6 questions; one further ∞ Bench-Sum task group, 1 question) to 1,074 questions over 14 task groups, against the full 3,671 questions over 29 tasks. All four competencies are covered; 17 tasks remain unsampled after the extension. The paired sample and its `entity2id` fingerprint are shared by all six runs (three paths $\times$ two generations); the 307 extension questions exist on Align+ only, and the two large extension document groups (1.9M and 5.7M characters) exceed the full-context reference’s strict budget of 1,042,576 characters, so reference coverage is 614 of the 1,074 questions. The reference uses the neutral prompt profile; the normalized-v1 profile of the same run scores 46.0 on its own 607-question coverage, a prompt-sensitivity artifact retained in the ledger and not used as a reference. Document-group lengths are heterogeneous and disclosed wherever cross-system cells lead: the two FactConsolidation document groups are the 6K variants (26,157 characters each); the 32K/64K/262K FC variants (136k–1.1M characters) and the other unsampled document groups were dropped for budget, all listed with context

RQ	Memory capability	Benchmark and observable prediction
1	Step-by-step source reading	MemoryAgentBench compares direct retrieval, repeated read-only memory calls, and source reading with citation checks on the same 767 questions. Scores should separate most on fact consolidation and multi-mention tasks when the agent can keep reading.
2	Temporal and multi-session access	LongMemEval tests updates and cross-session links. Versioned observations should keep both earlier and later facts available, while preference questions remain a boundary for latent-user-state modeling.
3	Relational consolidation	LoCoMo separates single-hop, multi-hop, temporal, and open-domain questions. Concept families and relation links should matter most when an answer joins episodes or revises an earlier value.
4	Revisable construction	Base/Align+ paired runs measure duplicate families, ingest calls, tokens, and task slices. Progressive alignment predicts lower construction cost and better consolidation without requiring a uniform score increase.
5	Deletion and absence boundary	MEME tests deletion, cascade, and absence. The current append-only store should retain source links but cannot yet guarantee suppression after deletion.

Table 4: Research questions and measurements. Each row tests a distinct prediction of the search-evidence separation; published comparison rows keep their test details so the score differences can be checked.

Arm	Alignment revision	Calls/doc	Total tokens/doc	Entity redirects	Duplicate groups
A	Baseline	68.4	145,378	0	8
B1	Window-batch alignment	24.2	108,467	1	12
B2	Batch review + final consistency check	29.4	111,853	245	6

Table 5: Three-arm alignment study on 137 LongMemEval-S documents per arm. B1 provides the large efficiency reduction; B2 spends about 3% more tokens than B1 on final consistency check and is the only arm with substantial non-destructive family redirection, reducing same-name duplicate groups. The quality slice is too small for a quality claim. B1 and B2 use the same saved service configuration; B2 differs through the runtime implementation and alignment switch.

sizes in the released ledger’s subset block; EventQA is sampled at the 64K/128K/full-book document groups rather than only the published full-book average; and the LRU domain carries $n=7$ paired and 14 extended questions (∞ Bench-Sum 1 then 2, DetectiveQA 6 then 12), so LRU cells are directional. The MAB source corpora are public; whether the answerer’s pretraining saw them cannot be excluded or measured post hoc, and every cross-system table row states its own answerer, judge, and subset.

A.4 ADJUSTED OVERALL DERIVATION

The paired sample omitted two tasks that score low for everyone, AR’s MH-QA and TTL’s Recommendation, and an earlier draft handled this by imputation: entering published non-full-context reference values (Recommendation ≈ 13 , MH-QA ≈ 60) gave an adjusted Overall of ≈ 59.0 against the raw 65.1 at the paired caliber, with a 59–61 range under sensitivity. The extension supersedes the imputation: both tasks are now measured directly on Align+ (source-reading MH-QA 77.0, Recommendation 59.8; Table 1), and the extended Overall is 71.0 with both tasks inside the aggregation. The extended sample’s composition still differs from the full benchmark’s (17 tasks remain unsampled), so we continue to state raw values only and never subtract across calibers; the imputed estimate is retained here as provenance for the released ledger.

A.5 PER-PATH PROMPTS AND CONFIGURATIONS

All paths use Kimi-K3 as answerer, with their prompts archived verbatim in the run manifests and code snapshots listed in Appendix A.1; the released supplementary material reproduces them in full. The Base→Align+ engine configurations differ as follows: max output tokens 16,384→32,768; usable context window 32k→262,144; retries 2→5; timeout 300 s→1,200 s; and absolute-path persistence enabled in Align+. Direct retrieval makes one fused retrieval call and one answer call; the Memory-tool agent uses a repeated read-only calls to the tool; the Source-reading agent materializes a document group, reads and verifies source files, and submits only spans that pass the citation check.

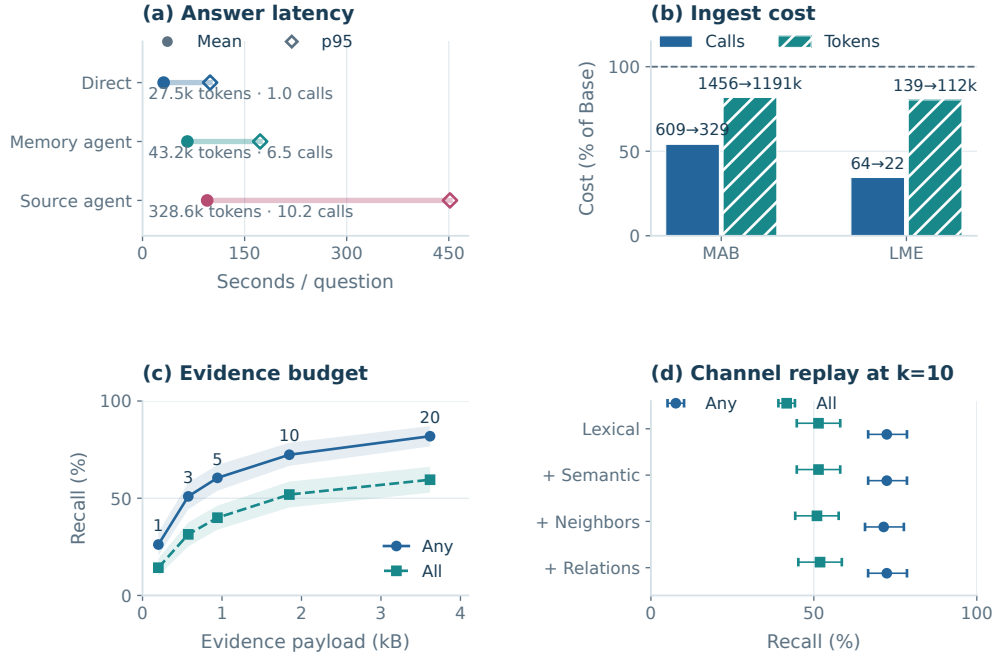


Figure 7: Operating-point diagnostics from frozen artifacts (Align+ engine; panel a over all 1,074 questions). (a) Mean-to-p95 answer latency with tokens and calls annotated; (b) paired Base→Align+ ingest cost; (c) the valid fixed-ranking retrieval-budget frontier; (d) source-only channel replay at $k=10$.

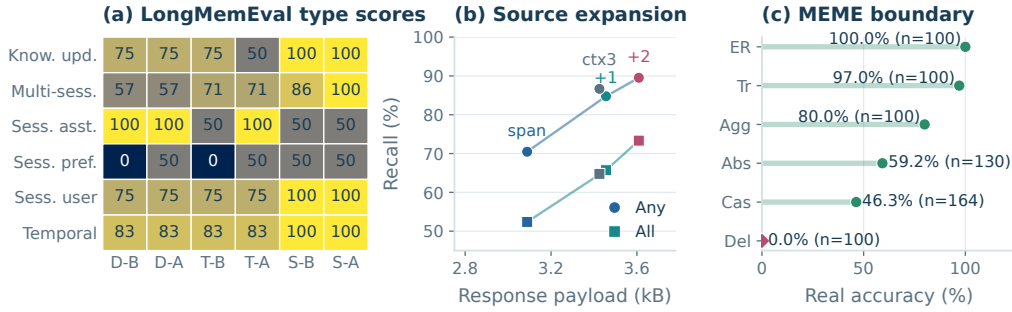


Figure 8: Additional quantitative diagnostics from the frozen ledger. (a) LongMemEval type scores for the paired Base/Align+ tracks; the directional samples have $n=25/25/27$ and are shown as coarse percentages. (b) Source-expansion recall versus response payload on 105 prior-error cases; this is a development diagnostic, not a held-out test. (c) MEME real task accuracy with task-specific denominators, exposing the 0% deletion boundary.

A.6 ADDITIONAL QUANTITATIVE DIAGNOSTICS

The ledger also contains type-level update scores, a controlled source-expansion replay, and the MEME deletion boundary. Figure 8 reports these quantities with their sample-size and development-set limits visible.

A.7 CARRIER ANCHORS (BIGCODEBENCH (ZHUO ET AL., 2025), ALFWORLD (SHRIDHAR ET AL., 2021))

BigCodeBench: 0.4859 completion / 0.2838 hard pass@1 (calibrated split). ALFWORLD: 0.9786 in-distribution / 0.9851 out-of-distribution success. Both are memory-insensitive agent-carrier

Table 6: X7 provenance ablation, offline retrieval replay on the 210-question gate set. Arms 1–4 form a clean channel ablation: only `enabled_channels` differs, so any recall delta is attributable to that channel set. Arms 5 (evidence-bound agent, gate ON) and 6 (unsurfaced evidence, gate relaxed) reuse arm 4’s five-channel ranking and are retrieval-identical to arm 4; their distinguishing columns (QA accuracy, unsupported-answer rate, mean evidence tokens) require the LLM agent run and are pending. The evidence-gate plumbing was proved offline: with the gate ON, submitting an unsurfaced turn is rejected; with the gate relaxed the same submission is accepted (`plumbing_ok=true`). In the deployed MAB runs the gate fired as designed: 14/1074 questions in the deployed *v2* skill-agent run had an evidence submission rejected as unsurfaced (9/767 in *v1*); 11 of the 14 resubmitted tool-surfaced evidence inside the loop and the remaining 3 ended at the step cap and were answered from tool-surfaced context; zero rejections on the LME runs. Bootstrap CIs: 2,000 resamples, 95%.

Arm	Channels (additive)	Recall-any@10 [95% CI]	Recall-all@10 [95% CI]	Mean bytes@10
1	raw-doc + episode-bm25	72.38 [66.67, 78.57]	51.43 [44.76, 58.10]	1833
2	+ semantic-provenance	72.38 [66.67, 78.57]	51.43 [44.76, 58.10]	1841
3	+ graph-neighbor	71.43 [65.70, 77.62]	50.95 [44.29, 57.62]	1845
4	+ relation-evidence (full 5)	72.38 [66.65, 78.57]	51.90 [45.24, 58.58]	1847

regressions: they anchor that the agent substrate executes tasks correctly, and are not compared horizontally with memory benchmarks.

A.8 MAB INGEST MANIFEST GAPS

The Base and Align+ MAB ingest runs share a 45-document paired intersection. Manifests missing from one generation are excluded from per-document efficiency ratios, which are therefore computed on the intersection only (609→329 calls/doc, −46%); token ratios use the same intersection. Question-level results are unaffected: all 767 paired questions run against frozen libraries in both generations, and the 307 extension questions ran on the Align+ engine only, so ingest-cost ratios remain defined on the paired intersection.

A.9 BUDGET FRONTIER

An earlier depth-10-versus-depth-20 replay is retained as an integrity finding: its prefix invariant failed on all 210 questions (`primary_result_invariant_violations=210`), so it is not used for any positive claim. The repaired design runs retrieval once per query at maximum depth and slices a fixed ranking to $k \in \{1, 3, 5, 10, 20\}$; prefix monotonicity then holds by construction (zero violations), and the frontier rises monotonically from $k=1$ to $k=20$ with per- k percentile-bootstrap CIs recomputed from the frozen replay artifacts.

A.10 X7 CHANNEL ABLATION DETAIL

A.11 PROTOCOL BOUNDARIES

The original LoCoMo token-F1 track includes 1,986 questions and five categories. The Mem0-compatible track contains 1,540 questions and four categories. LongMemEval-S contains 500 questions; the complete-write paired tracks use $n=25/25/27$ as described in Section 4.1. MEME’s 100 episodes produce heterogeneous task counts (ER/Agg/Tr/Del/Cas/Abs = 100/100/100/100/164/130 after-task checks in the cited run). Raw and real scores are not interchangeable.

A.12 INTEGRITY FAILURES RETAINED

The depth replay of Appendix A.9 reports `primary_result_invariant_violations=210` and `passed=false`. The MEME deletion result is 0% real accuracy. The X7 channel ablation is a null result, reported as an equivalence bound rather than dropped. The ∞ Bench-Sum $n=1$ flip costs the Align+ Overall 2.7 points directly (the net Base→Align+ gap is 1.8 points) and is disclosed at every comparison. The LME Direct retrieval–Memory-tool agent tie is reported although it is unflattering to the Memory-tool agent path. These records remain visible because an auditable memory paper should expose failed and null mechanisms as well as successful ones.

A.13 EXTERNAL COMPARISON TEST DETAILS

Mem0’s original paper reports LoCoMo category-level F1, BLEU-1, and a binary J in which a GPT-4o-mini judge labels each answer CORRECT or WRONG and the score is the percent judged CORRECT (mean±std over 10 independent runs). Zep’s original paper reports DMR and

Benchmark	Protocol	Score	Answerer / judge
LongMemEval-S (500)	sampled, $n=500$	84.60	Kimi-K3; Qwen3.7-plus cross-judge
LongMemEval, direct	paired, $n=25$	0.68 $\rightarrow$ 0.72	Kimi-K3 answerer
LongMemEval, tool-assisted	paired, $n=25$	0.68 $\rightarrow$ 0.72	Kimi-K3 answerer
LongMemEval, source-reading	paired, $n=27$	0.889 $\rightarrow$ 0.926	Kimi-K3; self-judged
LoCoMo-1540	Mem0 J scoring	95.19	Kimi-K3; answerer+judge; $n=1$
LoCoMo-1540 (same answers)	cross-judged	94.22	Qwen3.7-plus; agreement 98.51%

Table 7: Results on other benchmarks under their stated test procedures; each row states its answerer and judge. The LoCoMo run differs from Mem0’s published J in two admitted ways—single run versus 10-run mean $\pm$ std, and the answerer/judge model—and carries a disclosed legacy-fingerprint caveat; we state these conditions so the comparison is reproducible. The LME source-reading agent track’s two extra questions follow its session evidence rule; comparisons are made only within the same track.

LongMemEval with GPT-4o-family answerers. Rather than re-run those systems under our protocol, we reproduce their published tables verbatim below (Tables 8–10) as references reported under different test procedures. The Mem0 deterministic token-overlap metrics omitted from the main tables for space are reproduced in Table 12.

The K3-substitution row. The single Deep-Dream J row placed inside Mem0’s J column is produced by running Mem0’s exact binary scoring procedure with Kimi-K3 substituted for GPT-4o-mini as *both* the answerer and the judge—the only change is the model. This yields 95.19% under the same-model K3 judge (97.38/93.62/80.21/95.33 S/M/O/T) and 94.22% under an independent Qwen3.7-plus cross-judge on the same K3 answers (96.91/92.91/77.08/93.46; cross-judge agreement 98.51%). Two differences from Mem0’s published J are admitted: it is a single run rather than a 10-run mean $\pm$ std, and the model is Kimi-K3 rather than GPT-4o-mini. We additionally disclose an integrity caveat: the run manifest carries `diagnostic_override={kind:legacy-missing-runtime-code-fingerprint, formal_claim_allowed:false}`, because 1,529 of the 1,540 answer rows originate from a legacy source run (legacy-source-run) that predates runtime-code fingerprinting. The answers themselves are unchanged (`answer_rows_unchanged=true`, answer SHA-256 76af5772) and the per-item JSONL carries `runtime_code_sha256` on all 2,063 rows (3 distinct hashes: 2fa32774 on 2,019, 36b45933 on 33, 42afb1ac on 11), the same dataset SHA-256 (79fa87e9) and library snapshot (cf2e76b8) as the Qwen-answerer run, and the same Mem0 binary-judge prompt (repository `mem0ai/memory-benchmarks`, commit 4b61c5d, mode `unified-no-evidence-exact-direct`). We therefore report 95.19% as a diagnostic cross-judged result, not a fingerprint-complete formal entry.

A stricter controlled comparison—freezing the same answer model, judge model, prompt, temperature, evidence budget, dataset revision, and Graphiti/Zep implementation commit—remains the only path to a shared leaderboard, but it is not a submission hard gate now that the published numbers are cited directly.

A.14 AN INFORMATION-THEORETIC READING OF THE ARCHITECTURE

This appendix gives the paper’s two design commitments a formal shape. Every statement is either a structural property of the architecture (checkable against the schema and code) or a conditional proposition with explicit assumptions; none upgrades the causal wording discipline of Sections 4.2 and 4.8.

A.14.1 WRITE TIME: PROGRESSIVE ENTITY ALIGNMENT

Let the immutable source stream be $S = D_1, \dots, D_T$. Each extracted mention produces a feature vector X_t containing name and content embeddings, attributes, and relation-endpoint co-occurrences. Let Z be the latent entity assignment and let $M_t = (F_t, O_t, G_t)$ be the state after t observations: concept families F_t , an append-only observation store O_t , and a relation graph G_t . An observation is never replaced by a summary; it remains linked to its exact source span.

$$\begin{aligned}
 O_{t+1} &= O_t \cup \{(X_{t+1}, \text{span}_{t+1})\}, \\
 I(Z; O_{1:t+1}) &= I(Z; O_{1:t}) + I(Z; X_{t+1} \mid O_{1:t}) \geq I(Z; O_{1:t}).
 \end{aligned} \tag{4}$$

The second identity is a bookkeeping fact for an append-only stream: new observations can add information about identity, while no prior evidence is discarded. It explains why the system need not solve entity linking on the first mention. A low-confidence merge becomes an additional family (over-splitting); later observations can merge families by redirecting pointers while preserving both observation sets. A premature wrong merge is the dangerous error because it contaminates one search entry with two entities' evidence.

Proposition A.1 (Revisability). *In the append-only data model, any coarsening of the current family partition is realizable at a later time by non-destructive merges, with no loss of evidence. If a store instead overwrites an entity with a running summary σ , then $I(Z; \sigma) \leq I(Z; O)$ by the data processing inequality, with strict inequality whenever identity-discriminative detail is discarded.*

Proposition A.2 (Convergence under pooled evidence). *Assume features are i.i.d. conditional on entity identity and two distinct entities $e \neq e'$ emit distributions $P_e \neq P_{e'}$ with Chernoff information $C(P_e, P_{e'}) > 0$. For a same/different-entity test using k pooled observations, the likelihood-ratio error obeys $\Pr[\text{error}] \leq \exp[-kC(P_e, P_{e'})]$ (Chernoff, 1952). Hence the probability that a true merge, initially split under weak evidence, remains undetected falls exponentially with corroborating evidence.*

The assumptions are idealized: real windows are correlated, extraction noise is heavy-tailed, and same-name different-entity pairs can have small C . The prediction is therefore qualitative rather than a measured convergence rate. It matches the implementation's window-batch adjudication and final consistency check, which re-test candidate families with more pooled context. In the paired engine runs, same-name duplicate groups fall $340 \rightarrow 124$ (MAB) and $482 \rightarrow 99$ (LME), while ingest calls per document fall 46–65%; these are the measured reference values, not estimates of C . Relation edges serve two roles: endpoint co-occurrence helps distinguish hypotheses at write time, and the resulting graph supplies additional paths at read time.

A.14.2 READ TIME: HIERARCHICAL SEARCH AND STEP-BY-STEP READING

The derived layer is a search index, not an answer reconstruction. Let $A_t(q)$ map a question to concept families and relation neighborhoods, and let ρ be an agent policy that chooses which candidates to open. The resulting source contexts form an adaptive chain

$$C_0(q) \subseteq C_1(q) \subseteq \dots \subseteq S, \quad a_{k+1} = \rho(q, M_t, C_{\leq k}, u_k), \quad (5)$$

where u_k is the agent's current uncertainty. Concept search gives a fast first lookup; family observations and relation edges refine it; the final read returns the exact episode and span. This is why the same representation supports both autonomous exploration and source-reading answers.

Define reconstruction distortion as the discrepancy between cited text and the source text it claims to represent, and search loss as failure to place the relevant source span in C_k . Since every path resolves candidates to verbatim source spans and the gate accepts only spans surfaced in the current episode, reconstruction distortion is zero by construction; all residual loss is search loss. A summarizing memory fixes a rate-distortion point at write time, whereas ρ chooses the reading budget per query (Shannon, 1959). The agent can stop after a compact concept lookup, or disclose more neighborhoods and source files when uncertainty remains, without changing the authority of the evidence.

A.14.3 TWO-RATE DECOMPOSITION AND ANSWER RISK

Let $A(q)$ be the search record emitted by the derived graph index and let E_C be the exact source spans selected by a candidate set C . For the answer-relevant variable Y , the chain rule gives

$$I(Y; A, E_C \mid Q = q) = I(Y; A \mid Q = q) + I(Y; E_C \mid A, Q = q). \quad (6)$$

The first term is the information supplied by the compact concept/relation search record; the second is the information obtained by opening source text. A summary-only memory replaces S by a channel output $\tilde{S} = f(S)$ and cannot increase $I(Y; \tilde{S} \mid Q)$ beyond $I(Y; S \mid Q)$, whereas Deep-Dream leaves the source channel available after finding candidates. This is an information-bottleneck interpretation rather than a claim that the implementation solves the bottleneck variational problem (Tishby et al., 2000).

For completeness, define R_q as the event that a relevant source span is found and V_q as the event that the verifier answers correctly after reading it. Then

$$\Pr[\hat{Y} = Y \mid q] \geq \Pr[R_q \mid q] \Pr[V_q \mid R_q, q]. \quad (7)$$

The graph index primarily affects whether relevant text is found, while step-by-step source reading affects verification after the text is opened. The citation check prevents a citation to text that was never

Table 8: LoCoMo per-category J (binary CORRECT/WRONG accuracy, LLM-judged). Published rows reproduced verbatim from Chhikara et al. (2025) (arXiv:2504.19413, Table 1): Mem0 reports mean $\pm$ std over 10 runs with GPT-4o-mini as both answerer and judge (std omitted, all ≤ 0.75). **Deep-Dream** applies the binary J rubric pinned at repository commit 4b61c5d—a 2026 revision of the Mem0 judging prompt whose tolerance rules differ from the 2025 paper’s Appendix A—with Kimi-K3 as both answerer and judge (single run); the $\dagger$ row judges the same K3 answers with an independent Qwen3.7-plus cross-judge (cross-judge agreement 98.51%). Same 841/282/96/321 question populations (1,540 total, Mem0-compatible subset). *Caveat:* the K3-answerer run is flagged formal_claim_allowed=false—1,529 historical answer rows predate runtime-code fingerprinting (answers unchanged, sha 76af5772; per-item JSONL carries runtime_code_sha256); reported as a diagnostic cross-judge result, not a fingerprint-complete formal result. Deterministic F1/BLEU-1 omitted for space (Table 12). Weighted J , recomputed from the published category means under the 841/282/96/321 counts: Mem0 ≈ 62.14 , Mem0^g ≈ 61.36 ; the paper’s own Table 2 reports Overall $J = 66.88$ (Mem0) and 68.44 (Mem0^g). Deep-Dream agg. J : 95.19% / 94.22% ($\dagger$).

Method	J (S/M/O/T)
LoCoMo	— / — / — / —
ReadAgent	— / — / — / —
MemoryBank	— / — / — / —
MemGPT	— / — / — / —
A-Mem	— / — / — / —
A-Mem*	39.79 / 18.85 / 54.05 / 49.91
LangMem	62.23 / 47.92 / 71.12 / 23.43
Zep	61.70 / 41.35 / 76.60 / 49.31
OpenAI	63.79 / 42.92 / 62.29 / 21.71
Mem0	67.13 / 51.15 / 72.93 / 55.51
Mem0 ^g	65.71 / 47.19 / 75.71 / 58.13
Deep-Dream	97.38 / 93.62 / 80.21 / 95.33
Deep-Dream[†]	96.91 / 92.91 / 77.08 / 93.46

Table 9: Published Deep Memory Retrieval (DMR) baselines reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956). DMR has 500 multi-session conversations (5 sessions, ≤ 12 messages each) with one single-hop fact-retrieval question per conversation. Deep-Dream does not currently run DMR, so this table is a contextual reference only.

Method	Answerer	Accuracy
Recursive Summarization	GPT-4-turbo	35.3%
Conversation Summaries	GPT-4-turbo	78.6%
MemGPT	GPT-4-turbo	93.4%
Full-conversation	GPT-4-turbo	94.4%
Zep	GPT-4-turbo	94.8%
Conversation Summaries	GPT-4o-mini	88.0%
Full-conversation	GPT-4o-mini	98.0%
Zep	GPT-4o-mini	98.2%

opened; it does not guarantee semantic relevance. This factorization explains why improvements should be largest on questions requiring several mentions, temporal updates, or fact consolidation, and why a graph-only offline recall change need not predict the full online gain.

The measured operating points are (27.5k tokens, 0.333), (43.2k, 0.400), and (328.6k, 0.710) Overall on Align+ at the extended sample; the Base generation, frozen at the paired sample, replicates the ordering (0.3025/0.3759/0.6695). The FC-MH scores (2–4 versus 67–70) show where adaptive source reading matters most. X7 bounds the derived channels’ offline retrieval contribution at ± 2.4 points on the development slice; the graph index’s online value is document-group selection and link expansion, exercised by the source-reading agent rather than removed by the ablation.

Proposition A.3 (One-sided search error). *For any graph-index state, an error in A_t can change which spans enter C_k but cannot change the text or source identifier of a span admitted by the citation check. Thus a search error can miss relevant text, but it cannot fabricate a cited passage; the guarantee is that the cited source was opened, not that it entails the answer.*

This framing adds mathematical structure without upgrading the causal wording of the ladder: the exponential statement is conditional, duplicate-name counts are a proxy, and the measured ladder remains a systems comparison.

Table 10: LongMemEval main results. Published rows (GPT-4o-mini / GPT-4o, Full-context vs. Zep) reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956). The **Deep-Dream** row uses Qwen3.7-plus as both answerer and cross-judge—a different test procedure; its accuracy (84.6%) is shown beside the published numbers with that condition stated. Latency and average context for Deep-Dream were not measured under the published test procedure (—). The six-category breakdown is in Table 11.

Answerer	System	Accuracy	Latency (s)	Avg. context
GPT-4o-mini	Full-context	55.4%	31.30	115k
GPT-4o-mini	Zep	63.8%	3.20	1.6k
GPT-4o	Full-context	60.2%	28.90	115k
GPT-4o	Zep	71.2%	2.58	1.6k
Qwen3.7-plus	Deep-Dream	84.6%	—	—

Table 11: LongMemEval six-category breakdown (binary accuracy, %). Published Full-context and Zep values reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956); the **Deep-Dream (Q.)** row is Qwen3.7-plus answerer+cross-judge on the same six categories. Different answerer/judge, so this is a matched-granularity contextual reference, not a shared leaderboard. Sess. pref. is the hardest category for all systems.

System	Sess. pref.	Sess. asst.	Temporal	Multi-sess.	Knowledge upd.	Sess. user
Full-ctx (4o-mini)	30.0	81.8	36.5	40.6	76.9	81.4
Zep (4o-mini)	53.3	75.0	54.1	47.4	74.4	92.9
Full-ctx (4o)	20.0	94.6	45.1	44.3	78.2	81.4
Zep (4o)	56.7	80.4	62.4	57.9	83.3	92.9
Deep-Dream (Q.)	53.3	100.0	84.2	74.4	91.0	98.6

Table 12: Mem0 LoCoMo deterministic metrics, reproduced verbatim from Chhikara et al. (2025) (Table 1). F1 / BLEU-1 over the four categories (Single-hop / Multi-hop / Open-domain / Temporal). Deep-Dream does not compute token-overlap metrics (its metric is binary LLM-judge accuracy, Table 8), so no Deep-Dream column appears here; this table is included for completeness of the published reproduction only.

Method	F1 (S/M/O/T)	BLEU-1 (S/M/O/T)
LoCoMo	25.02 / 12.04 / 40.36 / 18.41	19.75 / 11.16 / 29.05 / 14.77
ReadAgent	9.15 / 5.31 / 9.67 / 12.60	6.48 / 5.12 / 7.66 / 8.87
MemoryBank	5.00 / 5.56 / 6.61 / 9.68	4.77 / 5.94 / 5.16 / 6.99
MemGPT	26.65 / 9.15 / 41.04 / 25.52	17.72 / 7.44 / 34.34 / 19.44
A-Mem	27.02 / 12.14 / 44.65 / 45.85	20.09 / 12.00 / 37.06 / 36.67
A-Mem*	20.76 / 9.22 / 33.34 / 35.40	14.90 / 8.81 / 27.58 / 31.08
LangMem	35.51 / 26.04 / 40.91 / 30.75	26.86 / 22.32 / 33.63 / 25.84
Zep	35.74 / 19.37 / 49.56 / 42.00	23.30 / 14.82 / 38.92 / 34.53
OpenAI	34.30 / 20.09 / 39.31 / 14.04	23.72 / 15.42 / 31.16 / 11.25
Mem0	38.72 / 28.64 / 47.65 / 48.93	27.13 / 21.58 / 38.72 / 40.51
<i>Mem0^g</i>	38.09 / 24.32 / 49.27 / 51.55	26.03 / 18.82 / 40.30 / 40.28